Ali Özkaya

Izmir, Türkiye | aliozkaya00@gmail.com | Portfolio
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Profile

Computer Engineering student who builds and ships LLM-powered systems end to end — data pipelines, REST APIs, fine-tuned models, and their evaluation. Comfortable across applied deep learning (PEFT, RAG, multimodal captioning) and the software engineering around it: Python services, containerization, automated testing, and reproducible pipelines.

Education

Izmir Institute of Technology (IZTECH) Expected 2026
B.Sc. in Computer Engineering — Izmir, Türkiye — GPA 3.08 / 4.00
Relevant coursework: Algorithm Analysis & Design, Theory of Computation, Operating Systems, Computer Networks, Computer Architecture, Parallel Programming, Probability & Statistics, Introduction to Machine Learning

Experience

AI Developer — IZTECH Software Society Sep 2024 – Present
Full-time — Izmir, Türkiye | github.com/raccoon-42/student-affairs-chatbot

- Built an Academic Information RAG chatbot answering Turkish-language questions about university academic calendars and regulations.
- Implemented hybrid retrieval (BM25 + dense semantic search) over a Qdrant vector store with multilingual E5 embeddings, plus PDF parsing and chunking pipelines.
- Served it via a FastAPI REST API and CLI (conversation history, pluggable OpenRouter / Ollama backends) and added an LLM-as-judge harness for response-quality evaluation.

AI Engineer — Research Ecosystems Jul – Sep 2025
R&D Internship — Izmir, Türkiye

- Built a natural-language-to-SQL service over academic data that converts free-form questions into validated, executable SQL queries.
- Designed an automated web research-and-extraction pipeline (OpenAI Agents SDK) that crawls sources, runs a multilingual NER model (XLM-RoBERTa, Hugging Face) to extract entities, and assembles structured academic profiles into a Neo4j knowledge graph and Qdrant vector store.
- Engineered the pipeline for reliability — event-driven architecture, typed structured outputs (Pydantic), error handling, and human-in-the-loop checkpoints.

Projects

Artist-Conditional Lyric Generation with QLoRA Adapter Blending 2026
Term project, CENG467 Natural Language Understanding & Generation | github.com/raccoon-42/lora-lyrics

- Trained per-artist QLoRA adapters (4-bit NF4, LoRA/DoRA) on Gemma 4 E4B for style-conditional lyric generation across five progressive-metal artists, locally on a 32 GB RTX 5090.
- Built inference-time **adapter blending** to interpolate between artist styles without retraining.
- Designed a three-axis evaluation (classifier attribution, cross-artist perplexity, lexical coverage) and showed the axes disagree; introduced a TF-IDF **style-weighted loss** and characterized its sign-dependent effect.
- Trained a RoBERTa artist-attribution classifier (87.3% accuracy, 0.86 macro-F1; 5 classes) as the evaluation instrument.

Audio Captioning: Model & Encoder Ablation Study 2025–2026
Graduation project, CENG416 | github.com/raccoon-42/audio-caption

- Built a ClipCap-style music captioner on MusicCaps — a frozen audio encoder, a learned MLP projection to a prefix of pseudo-tokens, and a language-model decoder, trained in two stages (projection, then LM fine-tuning).

- Ran a two-axis ablation: language model (GPT-2 vs T5-base) under a fixed encoder, and audio encoder (general CLAP vs music-specialized MERT, MusicFM, MuQ / MuQ-MuLan) under a fixed GPT-2, with embeddings precomputed once per encoder.
- Added architectural and decoding ablations (prefix length, dropout, projection depth; beam search, nucleus sampling, repetition penalty) with Optuna LR search and a per-model multi-seed noise floor.
- Benchmarked against zero-shot audio-LMs (Audio Flamingo, Qwen2-Audio, Qwen3-Omni) and dedicated music captioners (LP-MusicCaps, BLAP), with empirical train/test leakage detection and clean held-out reporting.
- Evaluated with `aac-metrics` (BLEU, METEOR, CIDEr-D, SPICE, SPIDEr, FENSE; FENSE primary) plus a human- and LLM-judge pairwise A/B study scored by Fleiss' kappa.

Parayüz Chat Service May 2025 – Mar 2026

Fintech side project (10-person team), bank-statement analysis

- Built the conversational chat service for a bank-statement analysis product (LangGraph), owning prompt design and the routing logic that directs user queries to the right tools.
- Fine-tuned a NER model (PyTorch, Hugging Face) to label bank transactions and extract structured fields; routed LLM calls through OpenRouter.
- Delivered as part of a 10-person team.

MHRS MCP Server 2025

Model Context Protocol server for Turkey's national hospital appointment system | github.com/raccoon-42/mhrs-mcp

- Built an **MCP server** exposing MHRS (Ministry of Health appointment system) as tools Claude can call — search doctors/slots, book, modify, and cancel appointments from natural language.
- Implemented authenticated session handling and browser automation (Selenium, geckodriver) behind a modular client/service architecture.

Music Analytics Engine 2024

REST data pipeline with cloud deployment — prototype

- Integrated third-party REST APIs (Spotify) into MongoDB time-series schemas and deployed the service to **AWS** and **Google Cloud**.

Technical Skills

- **Languages:** Python
- **ML / DL:** PyTorch, Hugging Face Transformers & PEFT, QLoRA / LoRA / DoRA, RoBERTa, CLAP, spaCy, Optuna, NLTK; RAG, vector search, embeddings, knowledge graphs; OpenAI Agents SDK, LangGraph, NER, natural-language-to-SQL
- **Engineering:** REST APIs (FastAPI), Docker, MongoDB, event-driven architecture, automated testing (pytest), cloud deployment (AWS, GCP)
- **Tools:** Git, uv, Jupyter, Linux, Qdrant, Neo4j, Ollama, LaTeX, BeautifulSoup / Tavily (web scraping)
- **Focus areas:** language model fine-tuning, parameter-efficient adaptation, retrieval-augmented generation, agent / tool orchestration, building & shipping LLM services, evaluation & reproducibility
- **AI coding assistants:** Claude Code, Codex, Gemini (used together in daily workflow)

Certifications

The Complete Node.js Developer Course — Udemy 2021

Node.js, Express, MongoDB, REST APIs — completed all three course projects

Languages

Turkish (native) | English — C1 (Erasmus English Exam, 90/100, IZTECH, 2024)

Volunteering

International Student Volunteer — Broaden Your Horizons Oct 2023

Erasmus+ youth exchange (Malta) on environmental sustainability — collaborated with international peers on awareness and community initiatives.

References

Prof. Tuğkan Tuğlular — Department of Computer Engineering, Izmir Institute of Technology
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